

Bidirectional ClickHouse & Flat File Data Ingestion Tool

Name: Balam Venkata Sai Bharath

Email: saibharath282002@gmail.com

College: Vellore Institute of Technology, Vellore

Mobile: 7032943252

Result:

The screenshot shows the 'Bidirectional Data Ingestion Tool' interface. It has three main sections: 'Source Configuration', 'Target Configuration', and 'Status & Results'. In the 'Source Configuration' section, 'ClickHouse' is selected as the source. The fields for Host, Port, Database, User, and JWT Token are all set to their default values. A 'CONNECT' button is present. In the 'Target Configuration' section, 'Flat File' is selected as the target. The 'Output Filename' is 'output.csv'. A 'START INGESTION' button is visible. The 'Status & Results' section shows 'Ready to start'.

This screenshot shows the same tool interface but with different configurations. In the 'Source Configuration' section, 'Flat File' is selected as the source. The 'Select File' field shows 'Choose File' and 'No file chosen'. A 'LOAD COLUMNS' button is next to the 'Delimiter' field. In the 'Target Configuration' section, 'ClickHouse' is selected as the target. The fields for Host, Port, Database, User, and JWT Token are all set to their default values. The 'Table Name' field is empty, and the 'Create new table' checkbox is checked. A 'START INGESTION' button is visible. The 'Status & Results' section shows 'Ready to start'.

Executive Summary

This document provides a complete technical and functional overview of the Bidirectional ClickHouse & Flat File Data Ingestion Tool. The application serves as a robust solution for seamless data exchange between ClickHouse databases and flat files (CSV/TXT), featuring JWT authentication, schema discovery, and efficient batch processing capabilities.

Objective

The Bidirectional ClickHouse & Flat File Data Ingestion Tool is a web-based application designed to facilitate seamless and secure data transfer between ClickHouse databases and flat files (CSV, TXT, etc.). It supports:

- **ClickHouse → Flat File (Export)**
- **Flat File → ClickHouse (Import)**

The tool empowers users with schema discovery, column selection, JWT-based secure authentication, real-time progress feedback, and robust error handling for both directions of data flow.

Objectives

- Develop a web-based interface for bidirectional data transfer
- Implement secure JWT authentication for ClickHouse connections
- Provide schema discovery and selective column transfer
- Ensure reliable error handling and user feedback
- Support large dataset processing with efficiency

Features Implemented

Core Functionalities

Bidirectional Data Flow:

- Export ClickHouse tables to flat files.
- Import CSV files into ClickHouse.

Interactive User Interface:

- Source and target selection (ClickHouse or Flat File).
- Dynamic input fields for connection parameters and file settings.
- Column selection with checkboxes.
- Action buttons: Connect, Load Columns, Preview, Start Ingestion.

Authentication:

- JWT-based token authentication for ClickHouse.
- Secure integration via clickhouse-driver and PyJWT.

Error Handling:

- Covers connection issues, invalid tokens, malformed files, and data mismatches.
- User-friendly error messages displayed in the UI.

Data Processing:

- Efficient batch processing.
- Records count tracking and display upon completion.

Technical Architecture

Backend

Language: Python

Framework: Flask

Libraries Used:

- **clickhouse-driver** – For ClickHouse interaction.
- **PyJWT** – For handling JWT-based authentication.
- **python-dotenv** – For environment configuration.
- **pandas** – For efficient file parsing and data transformations.

Frontend

- HTML/CSS/JavaScript (Vanilla)
- Bootstrap 5 – For layout and responsive design.
- Dynamic components for real-time updates (e.g., preview, status reporting).

Database

ClickHouse – Tested with local Docker-based instance using example datasets:

- uk_price_paid
- ontime

Setup & Configuration

Prerequisites

- Python 3.7+
- ClickHouse server (local or Docker)

Installation

Clone the repository:

```
git clone https://github.com/yourusername/clickhouse-file-ingestor.git
```

```
cd clickhouse-file-ingestor
```

Install dependencies:

```
pip install -r requirements.txt
```

Configure .env file:

```
CLICKHOUSE_HOST=localhost
```

```
CLICKHOUSE_PORT=8123
```

```
CLICKHOUSE_USER=default
```

```
CLICKHOUSE_PASSWORD=
```

```
CLICKHOUSE_DATABASE=default
```

Run the application:

```
python app.py
```

Access the UI at:

<http://localhost:5000>

How to Use the Application

◆ Step 1: Select Source

- **ClickHouse:** Enter host, port, DB, user, and JWT token.
- **Flat File:** Upload the CSV/TXT file.

◆ Step 2: Configure Target

- **ClickHouse:** Specify destination table name.
- **Flat File:** Set output file name.

◆ Step 3: Select Columns

- Fetch and display available columns from the source.
- Select desired columns using checkboxes.

◆ Step 4: Start Ingestion

- Click *Start Ingestion*.
- Monitor status updates and final record count.
- Download output file if exporting.

Testing & Validation

Test Cases Executed

Test Case	Status
ClickHouse → CSV (Single Table)	✓ Pass
CSV → ClickHouse (New Table)	✓ Pass
JWT Authentication Test	✓ Pass

Test Case	Status
Error Handling (Invalid File)	✔ Pass

Conclusion

The **Bidirectional ClickHouse & Flat File Data Ingestion Tool** delivers a reliable, secure, and intuitive interface for handling data movement between modern databases and flat file systems. With support for JWT-based authentication, schema discovery, data previews, and error handling, it streamlines the ingestion process for analysts, engineers, and data teams alike.